Review

Future ONCOLOGY

Regulatory role of CD39 and CD73 in tumor immunity

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CD39 is the rate-limiting enzyme for the molecular signal cascade leading to the generation of ADP and adenosine monophosphate (AMP). In conjunction with CD73, CD39 converts adenosine triphosphate (ATP) to ADP and AMP, which leads to the accumulation of immunosuppressive adenosine in the tumor microenvironment. This review focuses on the role of CD39 and CD73 in immune response and malignant progression, including the expression of CD39 within the tumor microenvironment and its relationship to immune effector cells, and its role in antigen presentation. The role of CD39- and CD73-targeting therapeutics and cancer-directed clinical trials investigating CD39 modulation are also explored.

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Immune checkpoint inhibitors have revolutionized oncology care over the last 10 years. However, while robust and durable responses have been demonstrated, a large subset of patients do not respond, and most of those who do in time demonstrate recurrent disease. Significant effort has been undertaken to identify mechanisms of immune resistance, biomarkers of response and additional immune response-modulating targets to improve therapeutic benefit.

Recently, CD39 has been identified as a critical immune-related target [1]. CD39, also known as ENTPD1, converts extracellular adenosine triphosphate (eATP) to adenosine (ADO). Increased ATP levels in the tumor microenvironment (TME) are associated with proinflammatory activity and ADO increases are anti-inflammatory. The balance of ATP and ADO in the TME guides the cancer immune response to immune therapy [2]. Other molecular signals, notably CD73, are also involved in this process. eATP is released by stressed, injured or dying cells and in response to hypoxic conditions of the intratumor microenvironment [1,3–5]. Extracellular ADO (eADO) is a known inhibitor of antitumor T-lymphocytes, which is highlighted by frequent overexpression of CD39 on the surface of malignant cells [6–8]. Given that CD39 plays such a critical role in the immunosuppressive ADO signaling network of cancer protection, there has been robust activity exploring therapeutic modulation of CD39 and CD73 expression and activity.

Overexpression of CD73 is seen in hypoxic conditions and enables a greater migration of individual tumor cells [5,9–17]. The neoplastic axis of CD39/CD73 can be summarized as such: ATP is released in the TME and quickly converted to ADO, which directly promotes cancer cell growth by enhancing the invasive and metastatic properties of the cells [13]. Among these is the engagement of ADO receptors on endothelial cells that enhances the production of proangiogenic factors such as VEGF, IL-8 and b-FGF [14]. In addition to enhancing the metastatic properties of tumor cells, CD39/CD73 activation also creates an immunosuppressive effect on macrophages, neutrophils, dendritic cells and T cells [11]. Both scenarios enhance the progression of malignant disease thereby supporting research exploring CD39/CD73 as biomarkers potentially related to clinical benefit and resistance [18]. More research into the correlation between CD39 expression and potential clinical biomarkers is needed, as expression and activation of CD39 also increase on all T-cell subsets associated with age [19]. In addition, there may

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be a correlation between genotype and CD39 expression. In particular, single nucleotide polymorphism (SNP) rs7096317 allows for CD39 expression on $CD4^+$ T-cells [20].

Upregulation of CD39 can be seen in multiple malignancies, including hepatocellular carcinoma, melanoma, leukemia and lung, gastric, colorectal, ovarian, breast and head and neck cancers [21]. Specifically, overexpression of CD39 is involved in the suppression of natural killer (NK) and T cells. Such results have been shown in lung cancer models [22]. Similarly, overexpression of CD73 is demonstrated in a variety of solid tumor malignancies, however, decreased expression is seen in genitourinary cancers, melanoma, breast cancer and cholangiocarcinoma.

It has also been shown that increased levels of tumor-expressive CD39 are correlated with a poor prognosis in numerous cancer types [23–26]. Increased CD39 is also associated with malignant progression and immune evasion [23,27]. Evidence of immune evasion was notably identified in ovarian cancer models. The clinical benefit of checkpoint inhibitors has been consistently lacking in numerous ovarian cancer studies [26]. CD73 plays a similar role by promoting cancer progression and metastasis through regulating the PI3K/AKT pathway and promotion of the Warburg effect [28]. Compared with the other facets of this signaling network, targeted therapeutics against CD39 and CD79 are just beginning to emerge in clinical trials. Currently, there are several products including monoclonal antibodies and nucleotide analogues that disrupt CD39 enzymatic activity and are undergoing early clinical testing [1,29,30]. These are discussed in subsequent sections.

Enzyme structure

CD39 is a 510 amino acid integral membrane protein with two transmembrane regions that dephosphorylate ATP in a Ca^{2+}/Mg^{2+}-dependent manner to ADO monophosphate (AMP). The active site of CD39 is contained in the extracellular domain within five apyrase-conserved regions. Phosphate-binding motifs (DXG) are essential for the interaction of the substrate and CD39 during enzymatic activity and hydrolysis to produce monophosphates. Glycosylation of CD39 at seven potential sites plays an essential role in the targeting of CD39 to the membrane, protein folding and enzymatic activity [31]. CD73 is a 548 amino acid dimer anchored to the plasma membrane that then dephosphorylates AMP to ADO [12]. CD73 is anchored to the cellular membrane at the C terminus through glycosylated phosphatidylinositol. The dimer is found in both open and closed configurations resulting from structural domain rotation [32]. CD39 is expressed on numerous cell types within the TME, including but not limited to stromal cells, endothelial cells, NK cells, B cells, dendritic cells, Langerhans cells, mesangial cells, monocytes, macrophages, neutrophils and regulatory T cells ($CD4^+$, $Foxp3^+$) [11,31]. CD39 is expressed in a variety of tissues including the lung, thymus, spleen and placenta [7]. CD73 is also thought to be upregulated in times of stress and inflammation concurrently with CD39 and has been found in many of the same tissues [33].

ATP/ADO signaling

eATP is recognized as a cell–cell communication molecule acting via purinergic receptors (P2X7 receptor is extensively studied) and nucleotide-hydrolyzing ecto-enzymes such as CD39 [34,35]. Normal conditions are characterized by low levels of eATP, 10–100 nmol/l, compared with intracellular levels of about 5000 mmol/l [34,35]. ATP initiates inflammation and apoptosis through purinergic receptors. The P2X7 receptor is involved in innate and adaptive immune responses [36]. During the innate immune response, damage-associated molecular patterns or pathogen-associated molecular patterns activate pattern-recognition receptors like toll-like receptors (TLRs), which then induce ATP release [37].

ATP is required for efficient phagocytosis of apoptotic cells [37]. One of the effector functions of macrophages in innate immunity is phagocytosis and their chemotaxis is promoted by eATP [38]. Macrophages also release ATP in response to various stimuli including microbial, which in turn induces calcium signaling in unstimulated, neighboring cells, further promoting efficient phagocytosis. M1 macrophages, which are proinflammatory and secrete IL-12 and TNF-α among other inflammatory cytokines, express high levels of P2X4/P2X7 receptors. The purinergic receptor activation triggers calcium signaling in neighboring cells, promoting FcR-mediated killing and further cytokine signaling [39]. M1 macrophages also activate nitric oxide (NO) synthase generating NO [38]. It was shown that the role of eATP is not specific to a single pathogen and the phagocytic capacity of macrophages was significantly decreased in the absence of eATP and the antagonism of purinergic receptors P2X7 and P2X4. The same study also showed that the expression of CD39 in mesenchymal stem cells (MSCs) inhibited macrophage phagocytosis, which was reversed when MSCs were incubated with ARL-67516, an ectonucleotidase inhibitor [39]. Additionally, activated $CD39^+/CD73^+$ MSCs establish paracrine ADO signaling loops, which lead to an overall suppression of antitumoral responses [40].

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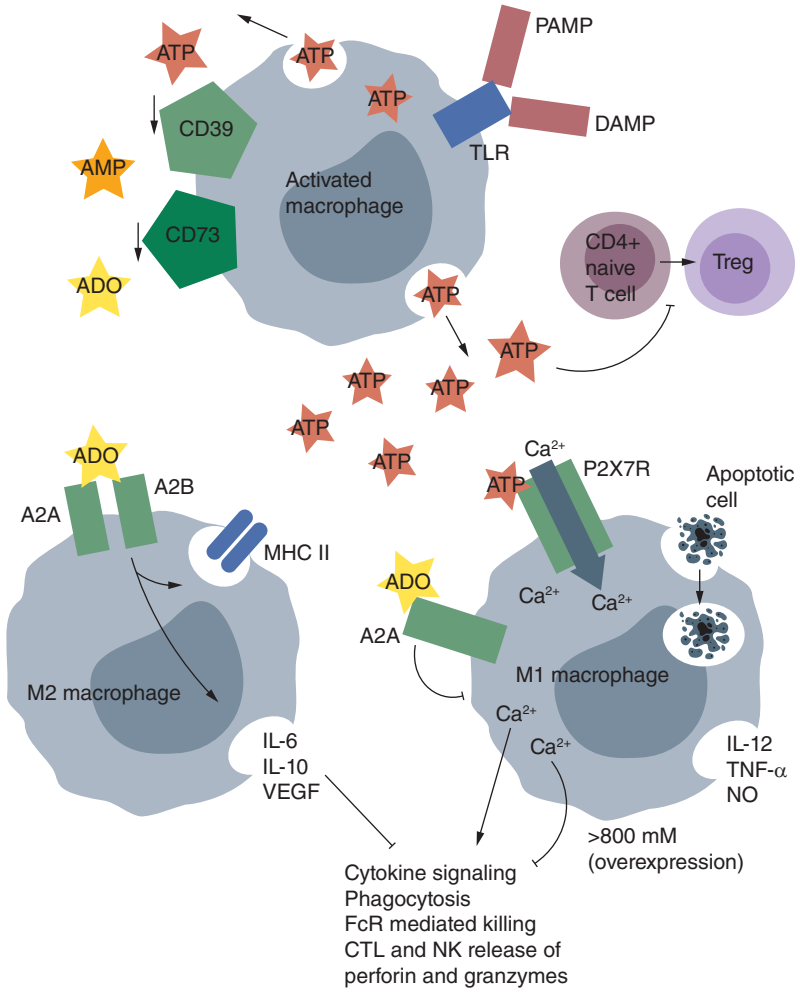

Figure 1. Activated macrophages (through microbial PAMP/DAMP interact with toll-like receptors, producing and releasing ATP into extracellular space. This eATP inhibits maturation of Tregs and can further macrophage differentiation into M1 macrophages by acting on P2X7 receptor. The same receptor allows for calcium influx, which at moderate levels promotes cytokine signaling; phagocytosis of apoptotic cells; release of inflammatory IL-12, TNF-α and nitric oxide and CTL and NK release of perforin and granzymes. Increased concentrations of intracellular calcium (>800 micromoles) can inhibit those same processes. On the other hand, eATP can be dephosphorylated by macrophage-bound CD39 and CD73 into eADO which furthers macrophage differentiation into M2 macrophages by acting on A2A/A2B receptors. This causes release of anti-inflammatory IL-6, IL-10 and VEGF, which serves to antagonize actions promoted by M1 macrophages.
ADO: Adenosine; AMP: Adenosine monophosphate; ATP: Adenosine triphosphate; DAMP: Damage associated molecular pattern; PAMP: Pathogen-associated molecular pattern; TLR: Toll like receptor; T-reg: T regulatory cell.

During adaptive immune response, ATP-P2X7 receptor signaling is necessary for T-cell receptor-mediated calcium influx, calcium wave propagation and IL-2 production [39]. Calcium acts as a second messenger mediating various physiological responses of neurons and apoptosis or cell death [41]. Cytotoxic T-lymphocytes (CTLs) and NK cells kill virally infected or cancer cells primarily by the release of cytotoxic or lytic granules containing perforin and granzymes at the immune synapses between them and the target cells. This mechanism is calcium-dependent [42]. ATP-P2X7 signaling also decreases the differentiation of naive $CD4^+$ cells into Tregs and decreases immunosuppressive activity (Figure 1) [36].

eADO has a half-life of a few seconds before it is catabolized to inosine by adenosine deaminase (ADA) and either transported intracellularly or becomes bound to G protein-coupled type 1 purinergic (P1) receptors. ADA controls the bioavailability of eADO and ADO signal propagation [36,43]. The P1 receptors are important for ADO signaling and include A1, A2A, A2B and A3. Signaling via A1 and A3 receptors inhibits the activity of adenylate cyclase and decreases intracellular cAMP levels, whereas signaling via A2A and A2B receptors do the

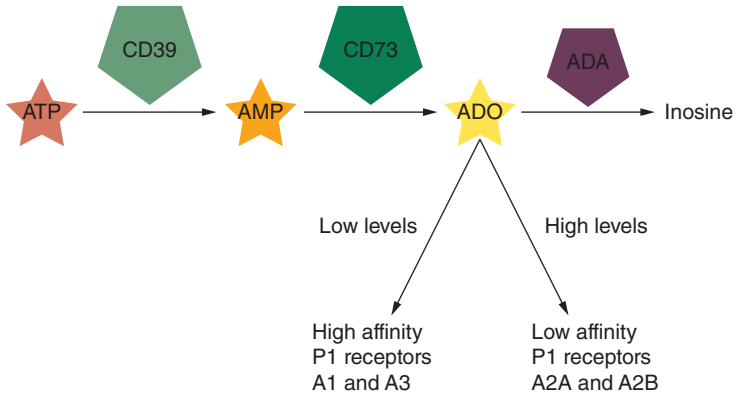

Figure 2. CD39 and CD73 act in synergy to dephosphorylate ATP into ADO. ADO can then act on either A1/3 or A2A/B P1 receptors depending on concentration. Low levels will trigger an inflammatory response. High levels will trigger an anti-inflammatory response.
ADO: Adenosine; AMP: Adenosine monophosphate; ATP: Adenosine triphosphate.

opposite and is primarily involved in promoting immunosuppression. Inosine, the product of ADO catabolism, can directly activate A2A, which triggers a different signaling pathway, ERK1/2, also contributing to the inhibition of proinflammatory cytokine production (Figure 2) [43].

CD39 receptors

The A2A receptor is a low-affinity receptor expressed on immune cells. Low-affinity binding occurs when there is a high concentration of a ligand and pathological conditions involve high concentrations of eADO [43]. Its activation stimulates macrophage differentiation and differentiation of $CD4^+$ T cells into Tregs, suppresses transcription factor NF-kB, blocking B-cell and TLR downstream effects and inhibits antigen presentation, counteracting T-cell receptor-mediated signaling and inhibiting production of inflammatory cytokines. The A2B receptor is also a low-affinity receptor expressed on myeloid cells and is therefore activated mostly under pathological conditions. Macrophages that upregulate A2A and A2B receptor expression are more likely to differentiate into M2 macrophages, which are anti-inflammatory in contrast to the proinflammatory M1 macrophages. M2 macrophages produce IL-10, VEGF and arginase 1, which all act to suppress the immune response [36]. A lack of CD39 in macrophages can therefore lead to ATP accumulation and over infiltration of inflammatory cytokines such as IL-6, IL-1B, IL-18 and TNF-α [11]. This explains why CD39 and CD73 are both decreased in proinflammatory M1 macrophages and increased in anti-inflammatory M2 macrophages [17].

In the early stages of inflammation and cellular stress, nanomolar concentrations of ADO bind higher affinity A1 and A3 receptors leading to enhanced chemotaxis toward inflammatory stimuli and FcR-mediated phagocytosis. Later in inflammation when ADO reaches micromolar (μM) concentrations, lower-affinity receptors like A2B and A2A are activated reducing neutrophilic chemotaxis, FcR-mediated phagocytosis and oxidative burst thus promoting anti-inflammation (Figure 3) [44].

CD39 & immunity

As previously described, ADO activates A2A and A2B, which serve to directly inhibit T cells of both markers [15]. CD39 positivity in $CD4^+$ T cells has been correlated with an increased sensitivity to apoptosis/metabolic stress and increased lymphocytic exhaustion [16]. CD39 positivity in $CD8^+$ T cells at baseline, on the other hand, has been associated with decreased levels of IL-2, TNF and IFN-g and, in addition, upregulation of inhibitory immune checkpoints [17]. Some of these inhibitory immune checkpoints that have been observed at increased levels in the presence of CD39-positive cells include but are not limited to LAG-3, PD-1, T-cell immunoglobulin and TIGIT and TIM-3. Enrichment of $CD39^+$ CD8 T cells has been noted in both primary and metastatic tumors [9].

Overexpression of CD39 and CD73 in $Foxp3^+$ Tregs enhances the concentration of ADO through two mechanisms: increased production and downregulation of ADA [45]. A2A receptor activation then upregulates the expression of PD-1 and CTLA-4 leading to the downstream effect of direct inhibition of CTL activation and response [17,46]. $CD39^+$ Th17 cells are therefore implicated in the suppression of overall immune responses to metastasis due to the ability to accumulate ADO [45]. CD39- and CD73-positive B-lymphocytes, in the same manner as T-lymphocytes, produce AMP and ADO and upregulate IL-6 and IL-10, which suppress T-cell mediated immune responses [47].

CD39 has also been implicated as a cell mediator of innate immunity, most notably with NK cells [46]. It has been shown that when CD39 is inhibited in tumor cells, a more potent and long-lasting antitumor response generated

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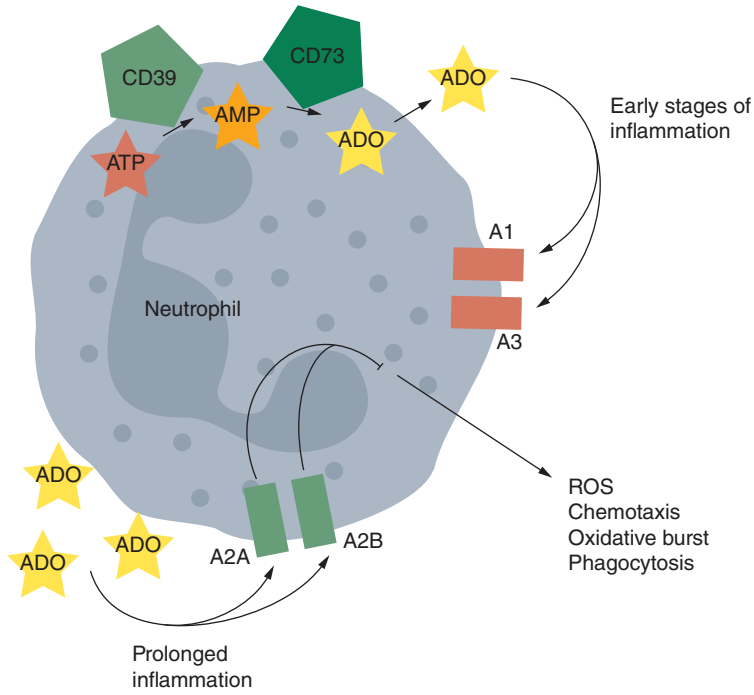

Figure 3. Neutrophil-bound CD39 and CD73 enzymes can potentiate chemotaxis, oxidative burst and phagocytosis through low levels of eADO at early stages of inflammation and inhibit the same processes during stages of prolonged inflammation when eADO levels rise.
ADO: Adenosine; AMP: Adenosine monophosphate; ATP: Adenosine triphosphate.

by NK and CD8$^+$ T-cell activity can be generated [48]. As mentioned, the immune response is attenuated by A2A receptor expression in NK cells [5]. This evidence again points to the critical balance between ATP and ADO, which is crucial for balancing the pro- versus anti-inflammatory environment in response to chronic metastatic disease. Dendritic cell immune response is also affected by ATP through the regulation of immunological synapses and signaling, which can result in immunosuppression via expression of the tryptophan metabolic enzyme indoleamine 2,3-dioxygenase and thrombospondin 1, which serve as negative proliferative T-cell regulators [22,25,40].

Tumor microenvironment

The TME is the extracellular space containing host cellular and noncellular components such as fibroblasts, endothelial cells, immune cells, cytokines, extracellular matrix and growth factors [49]. The TME composition determines the level of protection the tumor has against host immune responses. TGF-β, arginase, IL-10, VEGF, among other growth factors and anti-inflammatory cytokines accumulate in the TME promoting growth and invasion. The TME of many tumors (human ovarian carcinoma, human melanoma, mouse colon carcinoma and liver metastases of human colon carcinomas in mice) contain a high concentration of eATP, secreted by both host and tumor cells. CD39 and CD73 are upregulated in the hypoxic and acidic TME through transcription factors SP1 and hypoxia-inducible factor 1-alpha, serving an immunosuppressive role through eADO accumulation [50].

The mechanisms by which eATP and eADO exert their effects can be exploited in the TME to potentiate tumor growth. As previously discussed, low levels of eADO can aid in neutrophil chemotaxis and phagocytosis, while the accumulation of eADO acts to suppress the host's immunity [38]. Increased release of ATP extracellularly acts to increase phagocytosis by M1 macrophages and causes calcium influx for proper CTL and NK cell function. CTL and NK cells have low calcium optimum for efficient cancer cell elimination, which is between 23 and 625 μM. Moreover, at levels of about 800 μM, lytic granule release is reduced in CTLs [42]. Once calcium concentration surpasses 625 μM, CTL and NK cell functioning decreases and it becomes a delicate balance between eATP aided antitumor activity and accumulation of eADO from eATP degradation. The potentiation of calcium waves by eATP-P2X7 serves to surpass the optimal level necessary for CTL and NK cell activity and acts as a necessary substrate for cell proliferation in cancerous tissue [39]. Thus, there are two therapeutic opportunities involving CD39 and/or CD73. The first involves CD39 and CD73 antagonists to increase eATP, which would increase apoptosis. The second approach would be to upregulate CD39 and CD73 expression to decrease eATP production and decrease the accumulation of eADO, leading to an immune-stimulatory TME.

P2X7 receptor stimulation by eATP increases the production of reactive oxygen species associated with intracellular pathways affecting activation and release of tissue factor that is known to promote tumor progression, invasion and metastasis. Additionally, in the TME, ATP stimulates MMP-9 (a matrix metalloproteinase) release, which acts to degrade the extracellular matrix and thus facilitate tumor cell entry into blood circulation. For these cells to first enter the extravascular space and then survive at the metastatic site, ATP is necessary to ligate the P2Y2 receptor on the plasma membrane of endothelial cells to open the interendothelial junctions [50]. This receptor is also involved in stimulating IL-6-JAK1-STAT9 signaling, which promotes SOX9 expression mediating invasion and chemoresistance in breast cancer. The purinergic P2 receptors by which eATP exerts most of its effects are also overexpressed by almost all tumor cell lines that have been studied and this finding suggests a poor prognosis for the host [51].

Antitumoral immune responses & current/proposed therapies

Preclinical CD39 studies

As CD39 exhibits both ATPase and ADPase activity corresponding to a decrease in effective immune response (effective CTL activity) and by extension the opportunity for tumor cell progression, preclinical trials have focused on the concentration of CD39 as a potential marker for cancer progression. It has been demonstrated that CTLs with low levels of CD39 expression show higher anticancer cytotoxicity in human melanoma cells. The cells that overexpressed CD39 had lower malignant cell cytotoxicity [27]. Murine studies have shown that, in addition to blocking CD39 expression concerning CTLs, malignant CD39 inhibition can suppress tumor angiogenesis and is associated with decreased metastasis [21]. Suppressed growth in CD39-deficient tumors has been attributed to this suppressed angiogenesis in experimental models of B16-F10 melanoma, Lewis lung carcinoma and MC38 colon tumors in mice. This supports CD39 knockdown to control angiogenesis and improve the inflammatory response in the TME, reducing tumor growth [9,17].

Several studies have shown that inhibiting CD39 expression on the surface of malignant cells can be achieved with either CD39-blocking monoclonal antibodies (mAbs) or polyoxotungstate (POM-1), a broad inhibitor of ectonucleotidases. Overexpression of CD39 on malignant epithelial cells helps drive tumorigenesis by increasing ADO-mediated immunosuppression of NK cells and antitumor T cells [8]. Both anti-CD39 mAbs and POM-1 function to reverse this suppression of downstream proteins responsible for mounting cell-mediated immune responses. Reversal of suppression was found to affect immune cells that had previously infiltrated the tumor and newly recruited immune cells alike. One study compared the effectiveness of POM-1 versus 9-8B, an inhibitory mouse monoclonal anti-CD39 antibody that functions to block the enzymatic activity of CD39 in human cells [8]. The enzymatic inhibitory function of 9-8B was measured using flow-based platelet aggregation and radioactive CD39 cell-based assays, and results showed that both 9-8B and POM-1 significantly inhibited ADP-induced platelet aggregation. It was also shown that POM-1 was a more potent CD39 inhibitor, by about 46%, in $CD39^+$ fibrosarcoma cells compared with 9-8B [7,8,52–54].

POM-1 was shown to be effective at decreasing AMP generation in human myeloma cell lines and patient multiple myeloma cells via CD39 inhibition and increasing T-cell activation in combination with a CD73 antagonist *in vitro*. Despite this and findings that supported its potential efficacy through restricting metastasis in xenograft mice, POM-1 was also found to have increased toxicity toward nonmalignant cells due to broad ectonucleotidase inhibition in addition to a short therapeutic half-life [55,56]. Targeted therapy against CD39 using mAbs would eliminate the drawbacks of POM-1, likely limiting the future application of POM-1 in a clinical setting despite its potency [1,30,57]. Lastly, N^6-diethyl-β,γ-dibromomethylene-ATP (ARL67156) is an ATP analog that has shown promise as a CD39 competitive inhibitor *in vitro* [58]. However, its inability to function efficiently with high concentrations of ATP present *in vivo* in a murine model and in various assays *in vitro* implies that more developments could be made to optimize this treatment modality for effective clinical use [58,59].

In addition to studies investigating the efficacy of POM-1 and ARL67156 on CD39 inhibition, several preclinical studies investigating CD39-blocking mAbs *in vivo* have also utilized a tumor xenograft murine model. In one study, severe combined immunodeficiency disease (SCID) mice were grafted with either ARH-77 or MOLP-8 tumor cells, which were noted to have high levels of CD39 expression. After being treated with murine-derived human CD39-blocking mAbs at a concentration of 6 nM, ARH-77 xenograft mice were found to have a 41.7% reduction in growth 14 days post-treatment compared with the untreated control, while MOLP-8 xenograft mice were found to have a 32.43% reduction in growth 15 days post-treatment compared with the untreated control. Both reductions in tumor growth were significant [60]. A similar trend was observed in another study, in which xenograft

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mice received a combination of CD39-blocking antibodies and anti-PD-1 antibodies. The joint administration of CD39-blocking antibodies and anti-PD-1 antibodies appeared to have a synergistic effect, significantly decreasing tumor growth in the xenograft mice [30]. Another study utilized the CD39-blocking mAb, 9-8B, which was administered to NOG mice xenografted with CD39-overexpressing sarcoma tumors. Such mice were used to control against antibody-dependent cell-mediated cytotoxicity. Compared with IgG isotype control treatment, mice in the group treated with one dose of 6 nM 9-8B lived significantly longer, seeing an increased overall survival time of 21 days compared with the control group ($p < 0.0001$). While 100% of mice from the group receiving the IgG isotype control treatment perished by day 41 of the study, just 53% of the 9-8B-treated group perished by the same time point [8]. Importantly, CD39-blocking mAbs alone does not appear to be a definitive cure for $CD39^+$ tumors. However, this treatment modality is promising for its ability to significantly hinder $CD39^+$ tumor growth and extend the survival of subjects with $CD39^+$ tumors.

Key potential toxic effects of targeted CD39 inhibition have been identified in murine models. In one such study, knockout CD39-deficient mice were found to have impaired hemostasis due to abnormal platelet aggregation, and fibrin depositions in several organs [61]. This is likely due to the requirement of ADP for platelets in both primary and secondary aggregation, which is in part generated by the CD39 function [62]. Additionally, CD39-knockout mice develop liver cancer at an increased rate (70 vs 29%) due to the stimulated proliferation of hepatocytes. In *de novo*-induced CD39 knockout mice, tumor burden is also increased compared with control mice. The incidence of tumors over 5 mm in diameter was 69% in CD39 knockout compared with 6.7% in wild-type livers. This is believed to be caused by the constitutive activation of the Ras/MAPK and mTOR-S6K1 signaling pathways due to the abundance of eATP in CD39-null cells. The cells behave similarly to ATP-stimulated cells [63]. Thus, these consequences of CD39 inhibition must be carefully monitored during clinical trials. This also supports the previous conclusion that successful antitumor effects in acute and chronic disease can be achieved when ATP and ADO exert balanced pro- and anti-inflammatory effects.

CD39 as a biomarker

A recently published study identified high tissue expression of CD39 via NanoString analysis as a positive predictor of benefit from Vigil when administered to patients with newly diagnosed stage III/IV ovarian cancer undergoing maintenance therapy compared with similar patients who received Vigil with low tissue CD39 expression [18]. This study showed that CD39-high patients demonstrated improved overall survival (median not achieved vs 41.4 months; $p = 0.013$) and relapse-free survival (median not achieved vs 8.1 months; $p = 0.00007$) compared with CD39-low patients treated with Vigil. Vigil is a targeted immunotherapy functioning via multiple mechanisms. One of these involves the knockdown of cancer-expressed TGFβ1,2, which plays a role in the induction of CD39 and may contribute to a positive effect in Vigil-treated $CD39^+$ patients. Similar results have also been found by others. In one instance, high $CD39^+CD8^+$ T-cell proportion (determined by multiplex immunohistochemistry) was found to be a positive predictor of response to PD-1 or PD-L1 checkpoint blockade in non-small-cell lung cancer (NSCLC) [22,64], and in another study also involving patients with NSCLC, $CD39^+CD^+$ T-cells in peripheral blood was associated with longer progression-free survival and overall survival in patients treated with anti-PD-1 therapy [65]. These results, which outline a role for high CD39 expression as opposed to low expression, may be explained by the immune-suppressive effect related to the reduction of eATP in the TME. Vigil may preferentially overcome induction of the high CD39-suppressive TME, thereby conveying greater benefit within that subset of patients with ovarian cancer. Further testing of optimal Vigil-treated populations is underway and the mechanism of the high CD39 relationship to more robust benefit related to Vigil treatment. There are, however, conflicting preclinical and clinical results on the role of CD39/CD73 antagonists, discussed in later sections; these observations highlight the complexity of targeting the CD39/CD73 signaling axis. These studies suggest that CD39 expression may be justified for further study as a biomarker of sensitivity to immunotherapy.

Preclinical CD73 studies

CD73 also represents an attractive therapeutic target since it functions in the last step of AMP hydrolysis following the activity of CD39. Thus, CD73 inhibition could also be used to increase the recruitment of immune cells into the TME. Several preclinical studies have investigated the ability of CD73 antibodies to decrease cancer cell growth and metastasis. An anti-CD73 mAb, TY/23, was shown to significantly delay primary tumor growth of both 4T1.2 and E0771 murine breast cancer cell lines, with growth inhibition of over 50% between treated and untreated mice. There was also a significant reduction in the number of spontaneous metastases in treated mice compared

with untreated mice ($p < 0.05$). Immunocompromised SCID mice with 4T1.2 or E0771 tumors had no benefit compared with immunocompetent mice ($p > 0.05$ vs $p < 0.05$). These results support the requirement of an active adaptive antitumor immune response post-anti-CD73 mAb administration for the treatment to be effective. Researchers also found that the administration of TY/23 in mice irradiated and transplanted bone marrow cells from congeneic A2A-deficient mice was ineffective, also supporting the requirement for A2A for optimal response with anti-CD73 mAb therapy [66].

Clinical CD39 inhibitor trials

Presently, CD39-blocking mAbs are undergoing phase I clinical trials (Table 1). Four clinical trials are investigating the use of these antibodies as both monotherapy and as a component of combinational therapy. One study, NCT05075564, is investigating the safety and efficacy of recombinant human anti-CD39 IgG1 antibodies (ES002023) as monotherapy for the treatment of locally advanced and metastatic tumors (NCT05075564). Another study aims to determine the best-tolerated dosage of CD39-blocking antibodies (IPH5201) when used as monotherapy, or in combination with durvalumab or durvalumab + oleclumab to safely treat various advanced-stage solid tumors, with oleclumab being an anti-CD73 mAb (NCT04261075). An additional study is investigating the safety and efficacy of CD39-blocking antibodies (TTX-030) exclusively in combination therapy with pembrolizumab, budigalimab and/or chemotherapy in treating various advanced-stage solid tumors (NCT04306900). The basis for this study came from a preclinical study showing the efficacy of TTX-030 in reversing immunosuppression of the TME by reducing eADO, while also maintaining immunostimulatory levels of eATP [67]. Finally, the study NCT04336098 seeks to derive the safety and efficacy of CD39-blocking antibodies (SRF617) as both monotherapy and in combination therapy with gemcitabine + albumin-bound paclitaxel, or in combination therapy with pembrolizumab, in a dose-escalation model for the treatment of locally advanced-stage solid tumors and metastatic solid tumors. SRF617 administration was shown to successfully promote $CD8^+$ T-cell proliferation through successful CD39 inhibition *in vivo* within the microenvironments of orthoptic tumors. In this study, the pancreases of mice were implanted with tumor cells and treated with either SRF617 or a control antibody. The tumors were then examined 15 days after implantation and analyzed by flow cytometry to measure $CD8^+$ T-cell infiltration and SRF617 staining. The tumors showed strong SRF617 staining throughout the tumor while control antibody staining was present only in the necrotic core. Flow cytometry revealed that SRF617 induced a significantly higher $CD8^+$ T-cell tumor infiltration than the control antibody [68]. Though final results are yet to be published, each of the aforementioned clinical trials seeks to demonstrate the proinflammatory effect of CD39 inhibition leading to a more robust antitumor response by alteration of the TME *in vivo*.

Clinical CD73 inhibitor trials

There are currently two active clinical trials specifically investigating anti-CD73 mAbs. In a phase I study, both the maximum tolerated dose of anti-CD73 mAbs (IPH5301) as monotherapy and the recommended dose of IPH5301 in combination with chemotherapy and trastuzumab in treating advanced-stage tumors in patients with solid $HER2^+$ breast and gastric cancer are being investigated (NCT05143970) [69]. Another study is investigating the efficacy and safety of oleclumab (MEDI9447), an anti-CD73 mAb, in combination to treat advanced NSCLC with EGFR mutation (NCT03381274) [70]. At this time, results of NCT03381274 have been published and demonstrate both a tolerable safe profile and antitumor activity for participants receiving 80 mg oral osimertinib daily with either 1500 mg or 3000 mg oleclumab iv. (intravenous) every 2 weeks. Treatment-related adverse events (TRAEs) were reported in 60.0% of participants receiving the 1500 mg oleclumab treatment and 85.7% of the participants receiving the 3000 mg oleclumab treatment. Such reported events were generally mild with the most common including rash, stomatitis, diarrhea and paronychia. The researchers further note that all TRAEs reported by participants receiving the 1500 mg oleclumab treatment were related to osimertinib use only, while the majority (61.9%) of participants reporting TRAEs after receiving the 3000-mg oleclumab treatment experienced events related to oleclumab use. Grade 3 TRAEs were reported in 20% of participants receiving the 1500-mg oleclumab treatment and 14.3% of the participants receiving the 3000 mg oleclumab treatment and included hyponatremia, spinal cord compression, hemorrhage, pericardial effusion, pneumonia, cerebral infarction, liver abscess, respiratory arrest, pulmonary embolism, small intestinal obstruction, back pain, nausea and fatigue. No grade 4 TRAEs or deaths were reported during the study. Regarding the comparative effectiveness between the two treatment regiments, participants saw a disease control rate of 75.0% after receiving the 1500 mg oleclumab treatment and 82.4% after receiving the 3000 mg oleclumab treatment. Additionally, for patients who received the

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Table 1. Clinical trials.

Clinical trial title	Drug	End points	Results
A Study of ES002023 (Anti-CD39 Antibody) in Patients with Locally Advanced or Metastatic Solid Tumors NCT05075564	Recombinant human anti-CD39 IgG1 antibodies (ES002023)	Safety, tolerability, pharmacokinetics, pharmacodynamics and clinical activity of recombinant human anti-CD39 IgG1 antibodies (ES002023) as monotherapy for treatment of locally advanced and metastatic tumors	This study is ongoing and has no published preliminary data
IPH5201 as Monotherapy or in Combination with Durvalumab +/- Oleclumab in Subjects with Advanced Solid Tumors NCT04261075	CD39-blocking antibodies (IPH5201), oleclumab, an anti-CD73 mAb	Safety, tolerability, antitumor activity, pharmockinetics and immunogenicity of CD39-blocking antibodies (IPH5201) when used as monotherapy or in combination with durvalumab or durvalumab + oleclumab to safely treat various advanced-stage solid tumors, with oleclumab being an anti-CD73 mAb	IPH5201 was administered to 57 patients and was generally well-tolerated as monotherapy or when used in combination with dupilumab; 66.7% of patients has TRAEs, and the most common events were reactions to infusion (21.1%) and fatigue (17.5%). Median PFS was 1.4 months and median OS was 8.2 months; 22 (38%) patients had stable disease but no patient was observed to have a partial or complete response (www.esmoiotech.org/article/S2590-0188(22)00231-3/fulltext)
TTX-030 in Combination with Immunotherapy and/or Chemotherapy in Subjects with Advanced Cancers NCT04306900	CD39-blocking antibodies (TTX-030)	Safety, tolerability, pharmacokinetics, pharmacodynamics and antitumor activity of CD39-blocking antibodies (TTX-030) in combination therapy with various immunotherapies and/or chemotherapy in treating various advanced-stage solid tumors	This study is ongoing and has no published preliminary data
Study of SRF617 in Patients with Advanced Solid Tumors NCT04336098	CD39-blocking antibodies (SRF617)	Safety and efficacy of CD39-blocking antibodies (SRF617) as both monotherapy and in combination with gemcitabine + albumin-bound paclitaxel, or in combination therapy with pembrolizumab, in a dose-escalation model for treatment of locally advanced-stage solid tumors and metastatic solid tumors	SRF617 was administered to 55 patients (37 SRF617 monotherapy; 9 SRF617 + gemcitabine + albumin-bound paclitaxel; 9 SRF617 + pembrolizumab) and was generally well-tolerated. The most common TRAEs observed in each group were fatigue and nausea, with additional events including constipation, diarrhea and anemia having variable penetrance between each group, and were possibly related to drugs used in combination with SRF617. Interestingly, one patient with pancreatic cancer treated with SRF617 + gemcitabine + albumin-bound paclitaxel had a partial response (www.annalsofoncology.org/article/S0923-7534(21)04690-1/fulltext)
A Phase 1 First-in-Human Study of the Anti-CD73 IPH5301 Alone or in Combination with Chemotherapy and Trastuzumab in Patients with Advanced Solid Tumors NCT05143970	Anti-CD73 mAbs (IPH5301)	Maximum tolerated dose of anti-CD73 mAbs (IPH5301) as monotherapy, and recommended dose of IPH5301 in combination with chemotherapy and trastuzumab in treating advanced-stage tumors in solid HER2+ breast and gastric cancer	This study is ongoing and has no published preliminary data
Oleclumab (MEDI9447) Epidermal Growth Factor Receptor Mutant (EGFRm) Non-Small Cell Lung Cancer Novel Combination Study NCT03381274	oleclumab (MEDI9447), an anti-CD73 mAb	Safety, tolerability and antitumor activity of oleclumab (MEDI9447), an anti-CD73 mAb, in combination therapy with osimertinib to treat advanced NSCLC with epidermal growth factor receptor mutation	5 patients received oleclumab 1500 mg + osimertinib 80 mg and 21 received oleclumab 3000 mg + osimertinib 80 mg. Each regiment was generally well-tolerated. TRAEs were reported in 60.0% and 85.7% of patients from each group, respectively; 4 patients were T790M-positive and 3 of those were observed to have partial responses. Of T790M-negative patients, 25% receiving oleclumab 1500 mg and 11.7% receiving oleclumab 3000 mg were observed to have a partial response. Oleclumab 3000 mg regiment lead to a median PFS of 7.4 months and median OS of 24.8 months (https://pubmed.ncbi.nlm.nih.gov/36641093/)

mAb: Monoclonal antibody; OS: Overall survival; PFS: Progression free survival; TRAE: Treatment related adverse event.

3000 mg oleclumab treatment, the median progression-free survival was 7.4 months (95% CI: 3.6 months–not reached) and the median overall survival was 24.8 months (95% CI: 12.3 months–not reached). The performance of the 3000 mg oleclumab regiment as both a safe and effective immunotherapeutic agent has since prompted its use as the recommended phase II dose in subsequent clinical trials [69]. This finding is critical since advanced-stage

NSCLC is regularly able to generate resistance to the current standard of care, osmiertinib, which requires additional therapeutic agents in combination therapy to maximize positive outcomes in such cases [70].

Cotargeting CD39, CD73 & A2A

A recent preclinical study showed that inhibition of CD39 and CD73 using locked nucleic acid-modified antisense oligonucleotides (ASOs) inhibited ATP degradation and suppression of T-cell proliferation, which is seen with accumulation of ADO metabolites, CD39 was reduced by 98% when treated with 5 μM CD39 ASO and 95% with 2.5 μM CD39 + 2.5 μM CD73 ASO compared with the control. T cells that were treated with CD73 ASO alone showed a 70% reduction in CD73 expression compared with the control. Additionally, treated T cells were able to escape inhibition of proliferation and apoptosis induced by ADO metabolites. ASO-mediated knockdown of CD39 and CD73-induced suppression of proliferation was evident up to a concentration of 400 μM, which is comparable to what is measured at tumor sites [71]. Another study showed a synergistic effect of cotargeting CD73 and the ADO A2A receptor through antibody-directed therapies engaging Fcγ receptors, which can be of significant therapeutic benefit compared with single-target therapies. Gene-targeted mice deficient in both A2AR and CD73 had a reduction in tumor size: 15 mm^2 compared with 50 mm^2 in single gene knockout and 80 mm^2 in wild-type mice. This result was similarly observed with tumor mass: 50 mg compared with 100–150 mg in single gene knockout and 300 mg in wild-type mice. Additionally, 40% of dual-gene knockout mice rejected tumors compared with 7% in single-gene knockout mice. Additionally, A2AR inhibitor and anti-CD73 antibody activity were assessed in the 4T1.2 spontaneous metastasis model known to be responsive to either monotherapy. The result showed prolonged survival up to 80 days post inoculation compared with approximately 40 days in the control and 50 to 60 days in single therapy groups [72]. Although it was previously discussed that inosine may directly activate the A2A receptor leading to anti-inflammatory signaling through ERK1/2, Wang, *et al.* found that increased ADA2 expression is associated with improved prognosis in patients with multiple tumor types. This same study proposed PEGylated monomeric isoform of ADA as an anticancer immunotherapy as well as using ADA2 expression as a biomarker for patient selection [73].

Conclusion

Evasion of immune destruction is an emerging and evolving hallmark of cancer cells. CD39 expression has been identified in numerous cancer cell lines. Its role in immunosuppression is complex but may provide an opportunity for future cancer-targeting therapy in multiple areas of the signaling cascade. In addition to CD39-targeted therapies, other targets in this cascade are also being investigated, such as CD73, which may further enhance CD39 target activity. Clinical use as a biomarker of late-stage immune therapies (i.e., Vigil) may be another opportunity for medical management of cancer as our understanding of CD39 expands.

Future perspective

Many emerging clinical trials are showing hopeful results in targeting multiple parts of the neoplastic axis of CD39/CD73 with specific mAbs for the augmentation of treatment of many cancer types. Recently, high expression of CD39 has been shown to significantly predict positive responses in late-stage high-grade ovarian cancer to immunotherapy, which proves CD39 is a noteworthy biomarker. Not only are these combination therapies improving progression-free survival and overall survival, but they are also working to better treat symptoms such as cachexia. Other aspects of the same pathway, including the A2A and P2X7 receptors, are being investigated as potential combination treatment options in future clinical trials. Individualized cancer treatments have been building for almost a decade, with the US FDA first approving pembrolizumab in 2014 following the results of the KEYNOTE-001 clinical trial (NCT01295827) [71]. We envision that targeting the CD39/CD73 pathway and adjunct players will build on and revolutionize immunotherapy used across many cancer lines.

Author contributions

N Kaplinsky provided leadership, facilitated collaboration and wrote the manuscript. K Williams, D Watkins, M Adams and L Stanbery contributed to manuscript writing and editing. J Nemunaitis conceived the manuscript, oversaw manuscript preparation and provided edits.

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J Nemunaitis owns stock, is on the board of and is employed by Gradalis Inc. L Stanbery is employed by Gradalis, Inc. The authors have no other competing interests or relevant affiliations with any organization or entity with the subject matter or materials discussed in the manuscript apart from those disclosed.

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No writing assistance was utilized in the production of this manuscript.

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Executive summary

Enzyme structure
- CD39 is an integral membrane protein that dephosphorylates adenosine triphosphate (ATP) to adenosine monophosphate (AMP) and CD73 further dephosphorylates AMP to adenosine (ADO). These enzymes are expressed on different cell types within the tumor microenvironment.

Tumor microenvironment
- Tumor activity and host immune response hinge on the delicate balance between ATP and ADO accumulation within the tumor microenvironment and include the activation of receptors such as A2A and P2X7.

CD39 as a biomarker
- CD39 expression has been identified as a clinically significant positive predictor of ovarian cancer response to immunotherapy.

CD39 & CD73 inhibitor trials
- Anti-CD39 and anti-CD73 monoclonal antibodies are currently being investigated in clinical trials as monotherapies and in combination, with many underway.
- Oleclumab, an anti-CD73 monoclonal antibody, has recently been shown to be effective with a tolerable safety profile in treating non-small-cell lung cancer with EGFR mutation and will be moving to phase II (NCT03381274).

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